

HEALTH ECONOMICS (HE)

An Overview

Consider the following two proverbs:

- (i) "Health is wealth", and
- (ii) "Health is not everything in life, but without health life is nothing."

The second one indicates a dual property of health:

- health is a highly valued asset, and
- health is a prerequisite for other activities.

If "health is wealth", then Adam Smith for his classic book written in 1776, "An Inquiry into the Nature and Causes of the Wealth of Nations", became the father of modern economics can contribute equally to the study of the nature and causes of the health of nations. It has been historically manifested that healthy nations are wealthy and also that wealthy nations are healthy provided they allocate resources to improve the health of people through designing a feasible health care (HC) system. While the social reformers tell us that... "health is a right", the realization of that right is always less than complete because some of the resources that could be used for health and allocated to other purposes. No country is as healthy as it could be; no country does as much for the sick as it is technically capable of doing. The grim fact is that no nation is wealthy enough to avoid all avoidable deaths.

One may argue that unmet health needs may be filled with money that is "wasted" on cosmetics, luxury cars, cigarettes etc. But it is rather misleading because there are other goals, such

as justice, beauty and knowledge, which also remained unfulfilled because of resource limitations.

HE may thus be defined as a separate discipline in economics that applies the concepts, tools and theories of economics to the topics of health and health care (HC). It is primarily concerned with the issues related to making people as healthy as possible, given the social choices through implementing a HC system such that the "optimum level of health" is attained. For the health professionals the "optimum level" of health is the highest level technically attainable regardless of the costs, whereas economists are preoccupied with the "social optimum", which is defined as the point at which the value of additional investment of health exactly equals the costs of the resources required to obtain it. This is required for both resource allocation efficiency within the economy to the HC system and within the HC system to different activities and individuals. Also HC being a merit good* deserves distributional fairness in terms of equity considerations in financing and delivering HC services.

* Merit goods are those goods whose consumption is thought intrinsically worthwhile (to be contrastual with demerit goods, whose consumption is deemed harmful). Today, most societies provide for free public education and emergency health care; on the other hand, society also penalizes or forbids consumption of such harmful substances as cigarettes, alcohol, heroin.

What is health?

WHO (in 1958) defines... "health as a state of complete physical, mental and social well-being and not simply the absence of disease or infirmity."

Health, however, is not a tradable commodity (except may be for slavery); it is a characteristic of a commodity that we call HC, but then again, health is also a characteristic of seat belts, fire extinguisher, wholemeal bread, physical exercise, better environment and nutrition etc; health has only value-in-use, not much value-in-exchange.

Not so long ago, good health was considered to be a gift of God and poor health to be simply a curse of God or bad luck. The success of modern medicine have fastened the conviction that almost everyone can reach a good state of health, provided he/she uses the appropriate means. Often volatility in individual preferences for health are also observed; e.g., "sinning against health as long as health is good, sacrificing everything for health when health is bad." This makes health a state dependent phenomena.

Thus, health per se cannot be an object of economic analysis and our prime concern is with HC rather than health. Because crucial factors of health are money, and the provision of HC is only one; particularly, at lower level of income, other commodities such as food, shelter, clothing, safe water, hygienic sanitation, environment etc may be much more

significant.

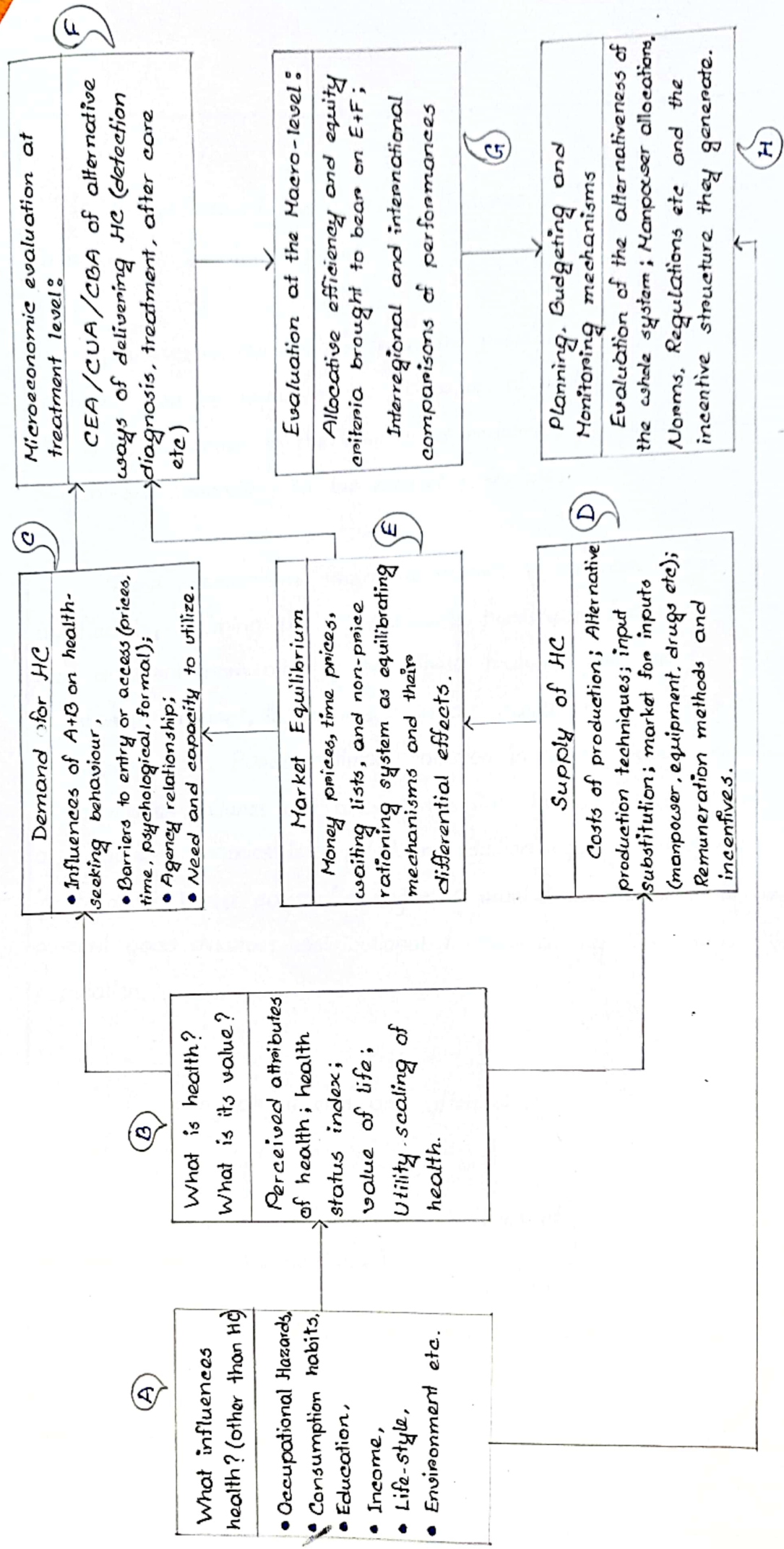
Other important aspect of HE is that it uncovers the cheerful face of a dismal science. Because we know that there are two things in life that are certain: Death and Taxes.

Dismal or gloomiest part of HE is that it is all about death, disease, pain and Taxes; cheerful part is that it is all about postponing death and reducing human sufferings, pain and taxes; and this essentially captures the main theme addressed in health economics (HE).

Moreover to understand the main elements in HE we may construct (a la Williams, 1986) a crude schematic chart that ties up the HC market components in terms of "Demand for HC" and "Supply of HC", including micro and macro evaluations of various health programmes.

[See chart-1 next page]

CHART-1: A Bind's Eye View of Health Economics as a separate discipline : Schematic Representation



Is HC different? Distinguishing features of HC services and their allocative implications.

Considerable deviations from the principles of a market economy can be found in the allocation of health care services. This is evident even in the developed countries, which are otherwise organized according to the market principles.

These deviations from the market system are generally justified by claiming that HC presents particular features rendering them different from other goods. These features are said to cause a "market failure", i.e., to result in an equilibrium that does not correspond to a Pareto optimal allocation in non-regulated markets. While market failures are present, the first fundamental theorem* of welfare economics is violated. In addition, equity consideration is adduced as a cause for rejecting market mechanism. HC being a merit good deserves distributional fairness among the target population.

Following arguments are often cited as the causes of market failure in HC:

(1) HC goods have the characteristics of public goods (non-rivalrous and non-excludable.)

(2) The consumption of HC services gives rise to external

* An equilibrium allocation achieved by a set of competitive markets will necessarily be Pareto efficient.

effects; both physical and psychological (altruistic) externalities arise.

(1) and (2) together suggests that mass vaccinations to prevent communicable diseases or measures designed to prevent epidemics, provided by the government and financed from general tax revenue, can lead to Pareto improvement.

(3) Option good property of HC services means that the existence of certain reserve capacity for extreme urgency constitute an option good, implies that mere existence of such goods yields utility to the consumers. HC provision may be treated as one type of option good.

(4) The production of HC goods and services is often characterized by improving returns to scale; it is a common perception in economics that perfect competition is not at all compatible with the increasing returns to scale.

(5) The prerequisite of a competitive market, i.e., market transparency and consumer sovereignty are not met because consumers lack enough information of rational decision-making who need to seek agents (physicians/doctors) for making appropriate HC choices, which is referred to as agency. Thus the absence of perfect information or rather asymmetric information causes HC market to fail.

(6) The existence of uncertainties in the incidence of disease and in the efficacy of diagnosis and treatment create special economic problems in HC. There is uncertainty involves in getting sick and receiving appropriate treatments; i.e., both sickness episodes and treatment regimen are all subject to variable risks and uncertainties.

(7) A rational risk-averse individual has high discount rate of future wants. In the absence of public interventions, the individual may resort to saving buying health insurance. However, since most individuals estimate future consumption as being less valuable than current consumption, many will simply fail to keep enough provision for the future. Consequently, means will be lacking in old age, resulting in a systematic under-provision of HC goods, when it is needed most.

(8) In the presence of both uncertainty and information asymmetry hinders the development of private health insurance market due to "adverse selection" and "moral hazards".

(9) Significant barriers to the mobility of resources exist because of limits on entry into health profession and by institutions into various institutional markets. Pharmaceutical industry is highly regulated through patient rights and quality controls.

(10) It is often argued that public attaches greater importance to equity than to efficiency in HC. It is also suggested that academics and policymakers agree less over what they mean by equity than they do over what they mean by efficiency. McLochlan and Hagnard (1982), for example, have suggested that, "... equity, like beauty, is in the eye of the beholder...." However, a broad agreement is that HC should be delivered according to need and ought to be financed according to ability to pay. The consideration of equity in the distribution of HC goods of services deserves particular attention among the population who cannot afford to pay.

The Role of Government in Health Care (HC):

Given the arguments of market failures in HC sector, there appears a *prima facie* reason for government intervention in the health industry. In fact, both efficiency and equity considerations motivate government policy.

Let us take the pure market solution where each individual pays 100 percent of its medical bills and there is no government programmes for the poor, older and underprivileged persons in the society. The free market can solve market failures due to adverse selection and moral hazards, but only at greater cost to individuals. Pure market does reduce inefficiency in the resource allocation but only at a greater cost in terms of equity consideration. It is therefore not surprising that few economists or politicians endorse the pure market model. Because the control of communicable diseases and the development of basic science and public goods that the market will not efficiently provide.

At the other extreme is a nationalized health care system (like NHS in the UK and Italy and NHI in Australia and Canada) in which HC is provided free of cost to all on an equal basis. Because HC is universal, there is no adverse selection and govt. can provide the public good of information and prevention. But increasing taxes often harm incentives to work and save, and moral hazard is too severe because consumers have few incentives to limit their use of services and the cost grows even more rapidly. The combination of price ceilings and subsidized services leads to chronic excess

demand for HC. Govt. also practices non-price rationing in the form of waiting for services. In other cases, health providers ration services by deciding which patient is most deserving.

An interesting example of rationing comes in wartime, when medics employ triage. This practice derives from the French producers in WWI of dividing the wounded into three categories:

- (i) those who would be helped by medical treatment;
- (ii) those who would survive without much attention; and
- (iii) those who were likely to die no matter how much medical attention they received.

Only first category received most medical services. Can you see why this hard-headed strategy is cost-effective in that it maximizes the number of lives saved? Such a strategy has been proposed by the World Bank in its 1993 WDR. The WB recommends that countries focus on those medical treatments that show the greatest increase in the number of "DALYs" per capita of expenditure. This approach is similar to that of consumers maximizing utility or firm maximizing profits. Some States of the United States have proposed rationing HC on a similar principle.

Applications of Supply and demand tools to HC sector

1. When govt. imposes price ceiling then price is not allowed to rise to balance supply and demand, some other mechanism must ensue to "clear the market."

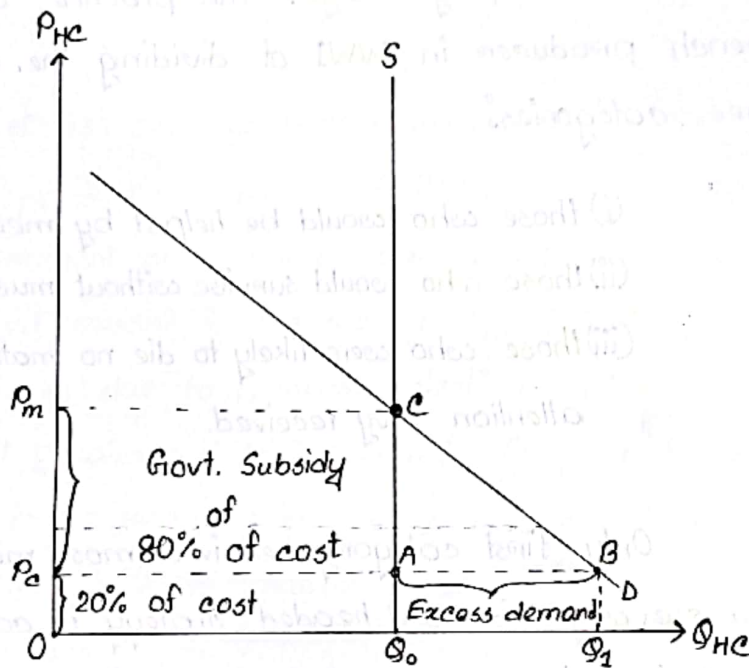


Figure-1: Rationing in HC market

Figure-1 illustrates non-price rationing in the HC market. Excess demand AB is checked off by some other mechanism such as long waiting list for elective surgeries or other non-emergency services.

2. The effect on price and medical services of an increase in demand when there are different supply elasticity:

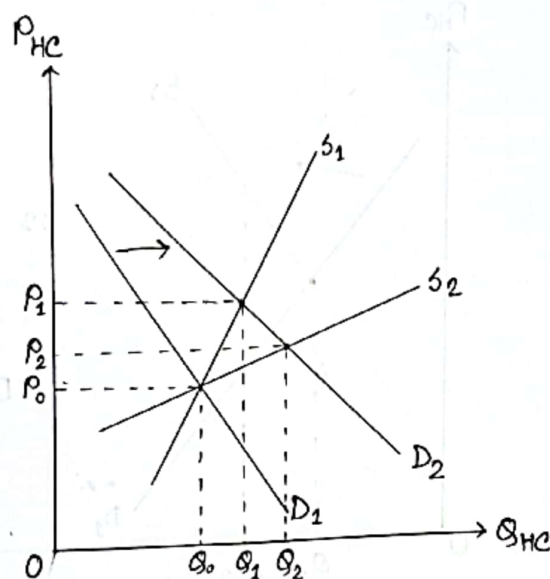


Figure - 02

In Figure-2, an increase in demand (D_1 to D_2) will affect P and Q to services differently, depending upon the elasticity of S . If S is relatively elastic (S_2), then an increase in demand from D_1 to D_2 will be accompanied by a price increase, but it will be much less (P_2) than if S were inelastic (S_1), in which case the new price would be P_1 . Similarly, a much greater increase in the quantity of services provided will occur under conditions of elastic supply (Q_2 vs Q_1). If the increase in demand was to occur as a result of a govt. subsidy programme, and if S was inelastic, the increased govt. expenditures for that programme would pay for higher prices (P_1) and less output (Q_1) of HC than if S was more elastic (S_2).

Thus, if markets with relatively inelastic supply can be made more elastic via various promotional campaign by the govt., the potential benefits in terms of lower prices and increase in HC services consumption may be very great.

3. Alternative D and S policies to achieve a redistribution of Health care:

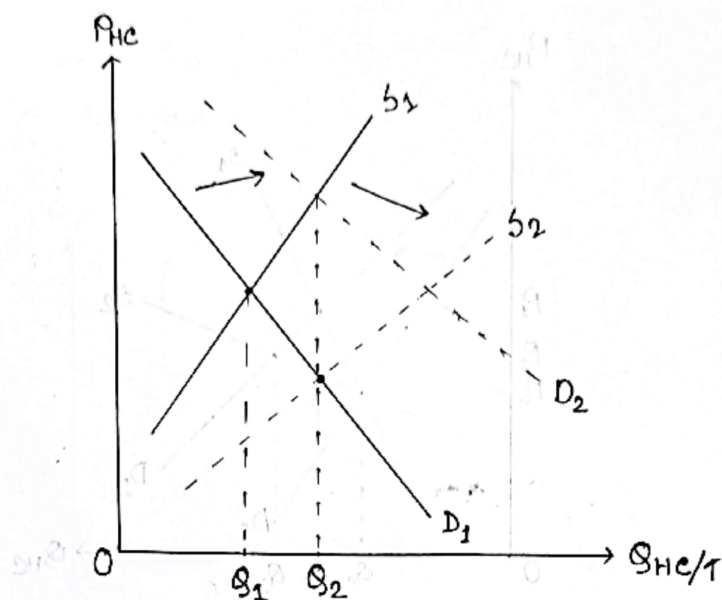


Figure - 03

To increase quantity of HC services consumed, as shown in Figure-3 (based on a normative judgement that it is desirable to do so), either D curve for HC can be increased from D_1 to D_2 , or the supply can be increased from S_1 to S_2 . Either of these policies will achieve the objective of increased case from Q_1 to Q_2 .

However, these alternative redistributive policies will differ in their costs and their effects on different population groups. Public policies designed to redistribute HC by means of supply subsidy and demand subsidy will have to be done based on both efficiency and equity considerations. The simple S and D tools, thus, enable health sector policymakers to address the issue of a desirable HC system.